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ECE – 782
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04/23/2026

LabTrack_UM Test Document

1. Introduction

The purpose of this document is to describe how the LabTrack_UM system was tested to make sure it works correctly. Testing was done to confirm that all main features function as expected.

2. Testing Approach

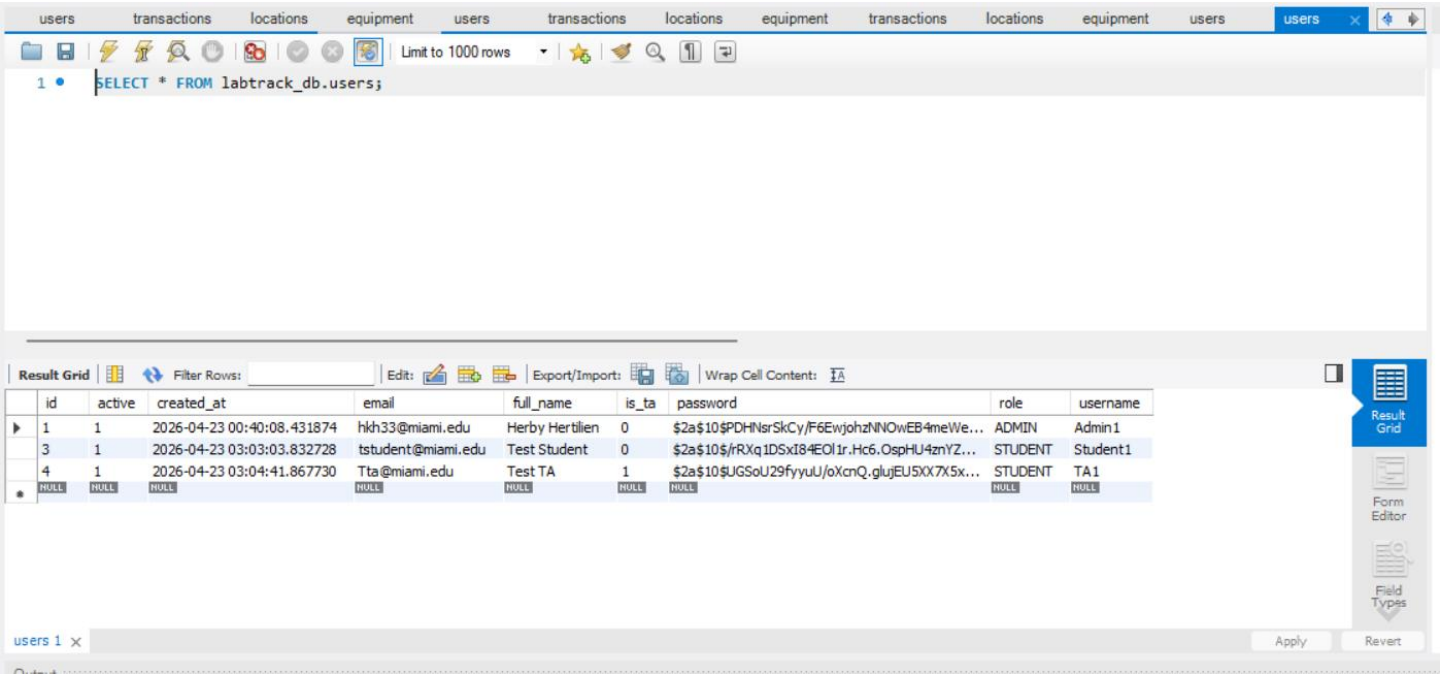
The system was tested by running the application and manually checking each feature. Each module was tested separately and then tested together as a complete system.

3. Test Cases

User Module:

- Created a new user
- Viewed existing users
- Updated user information

Expected: User data is saved and displayed correctly
Result: Passed

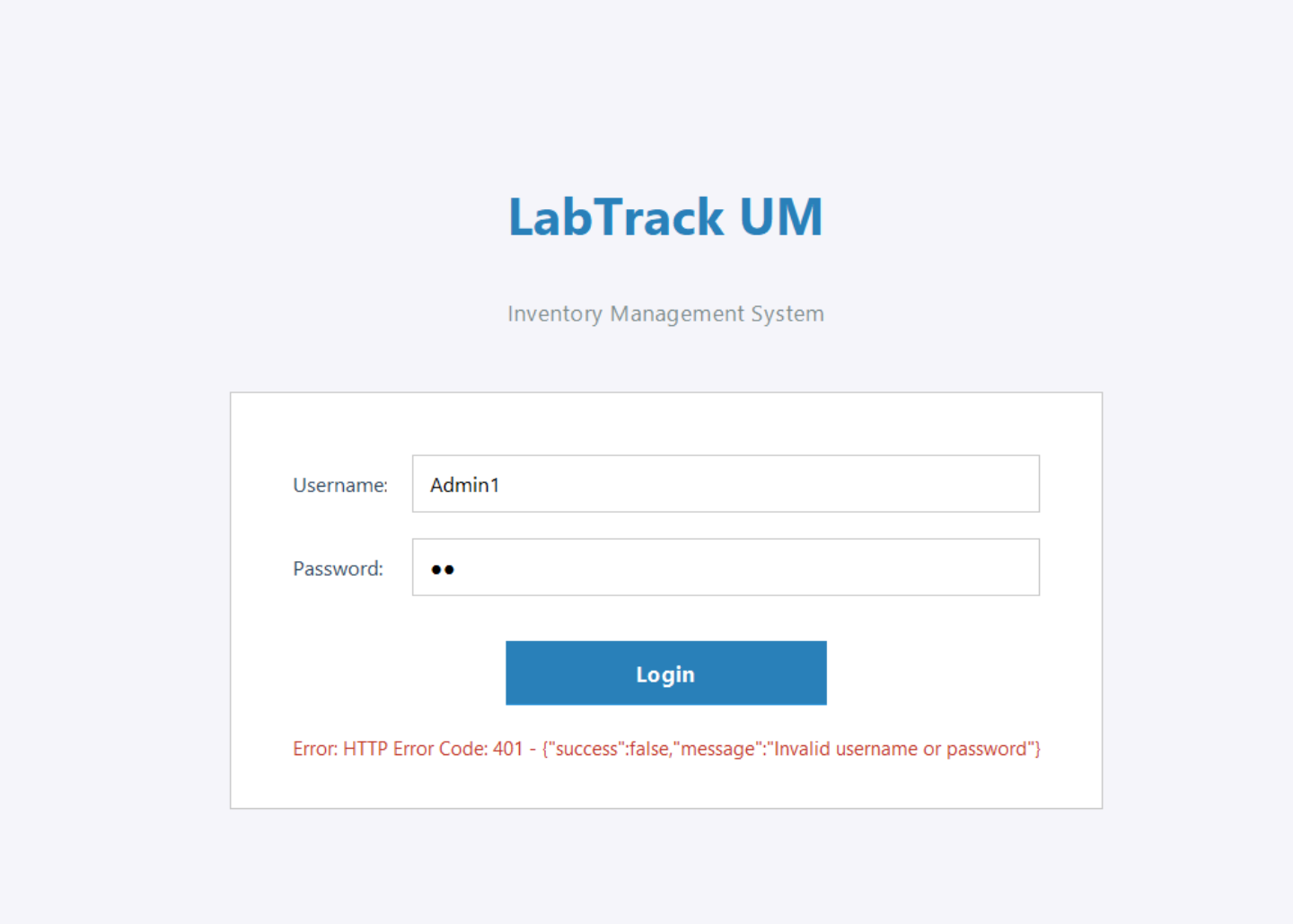


id	active	created_at	email	full_name	is_ta	password	role	username
1	1	2026-04-23 00:40:08.431874	hkh33@miami.edu	Herby Hertilien	0	$2a$10$PDHNsrSkCy/F6EwjohzNNOwEB4meWe...	ADMIN	Admin1
3	1	2026-04-23 03:03:03.832728	tstudent@miami.edu	Test Student	0	$2a$10$/rRXq1DSxI84EOl1r.Hc6.OspHU4znYZ...	STUDENT	Student1
4	1	2026-04-23 03:04:41.867730	Tta@miami.edu	Test TA	1	$2a$10$UGSoU29fyyuU/oXcnQ.glujEU5XX7X5x...	STUDENT	TA1
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Authentication Module:

- Logged in with valid credentials
- Attempted login with invalid credentials

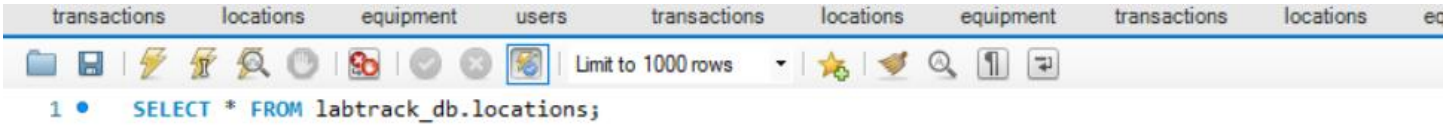
Expected: Valid login works, invalid login is rejected
Result: Passed



Location Module:

- Added a new location
- Viewed list of locations

Expected: Locations are stored and displayed correctly
Result: Passed

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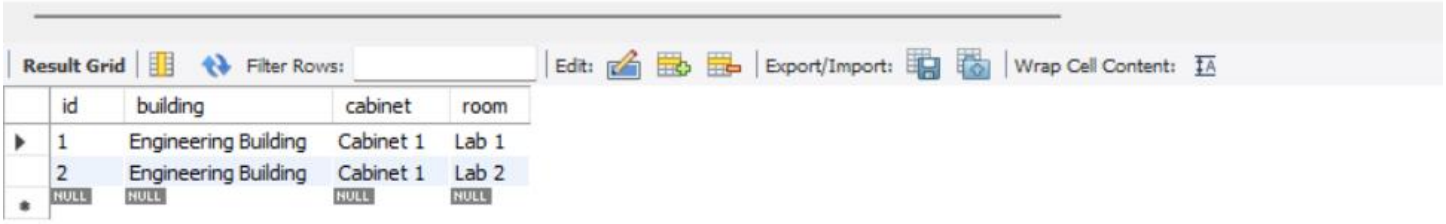
1 •  SELECT * FROM labtrack_db.locations;
```



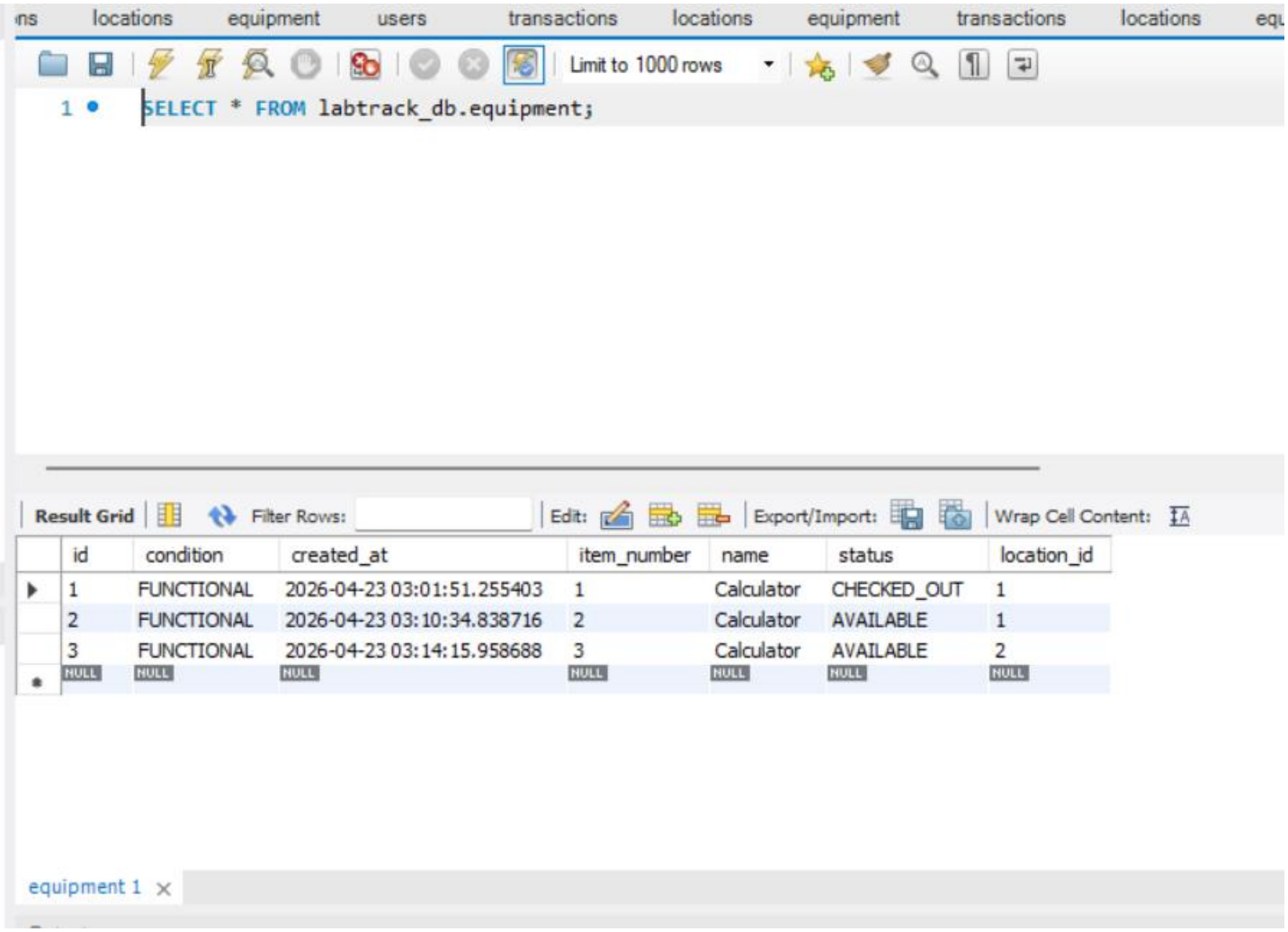
id	building	cabinet	room
1	Engineering Building	Cabinet 1	Lab 1
2	Engineering Building	Cabinet 1	Lab 2
NULL	NULL	NULL	NULL

locations 1 ×

Equipment Module:

- Added new equipment
- Updated equipment details

Expected: Equipment is saved and updated correctly
Result: Passed

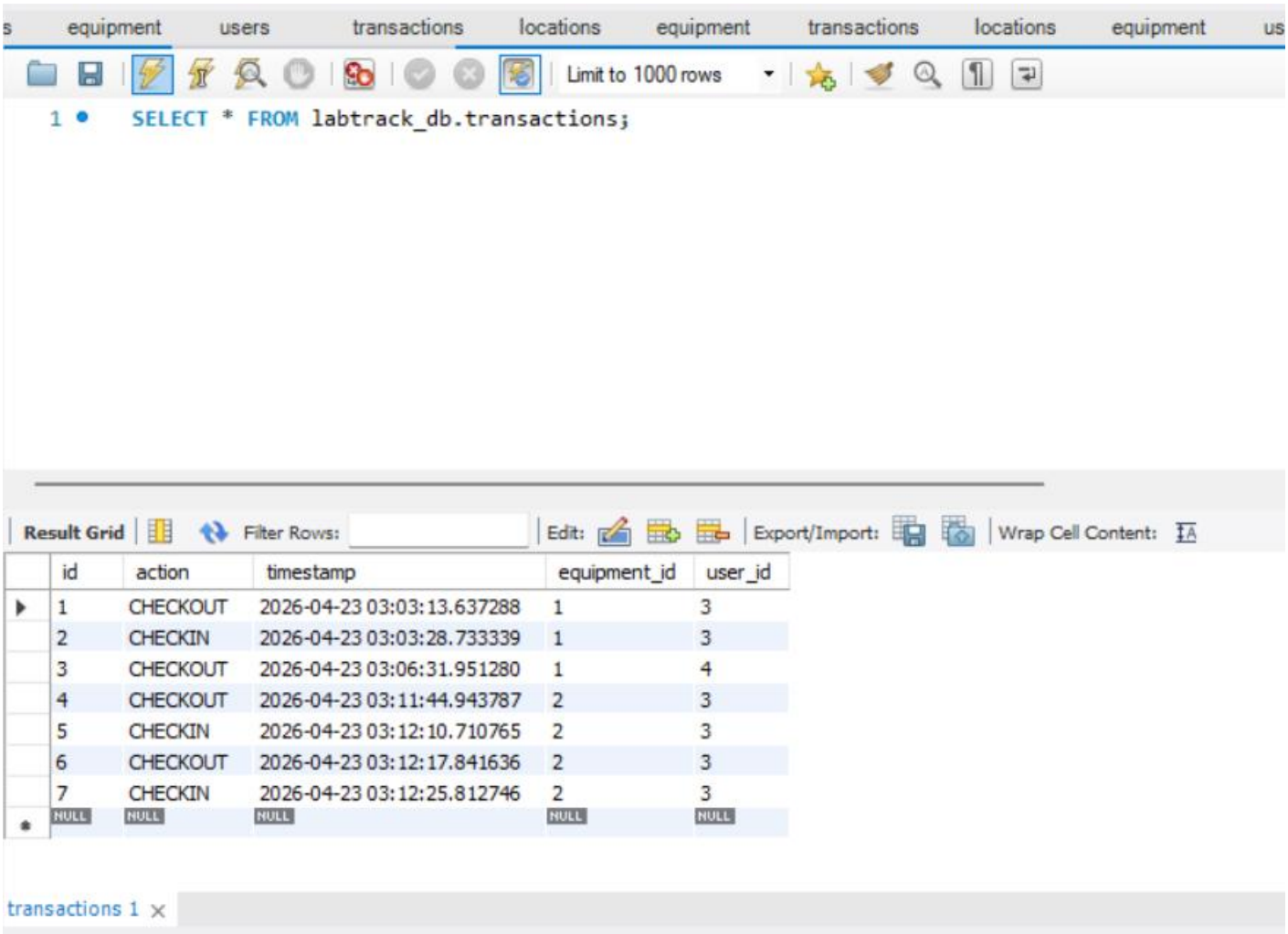


id	condition	created_at	item_number	name	status	location_id
1	FUNCTIONAL	2026-04-23 03:01:51.255403	1	Calculator	CHECKED_OUT	1
2	FUNCTIONAL	2026-04-23 03:10:34.838716	2	Calculator	AVAILABLE	1
3	FUNCTIONAL	2026-04-23 03:14:15.958688	3	Calculator	AVAILABLE	2
NULL	NULL	NULL	NULL	NULL	NULL	NULL

Transaction Module:

- Checked out equipment
- Checked in equipment

Expected: Transactions are recorded correctly
Result: Passed

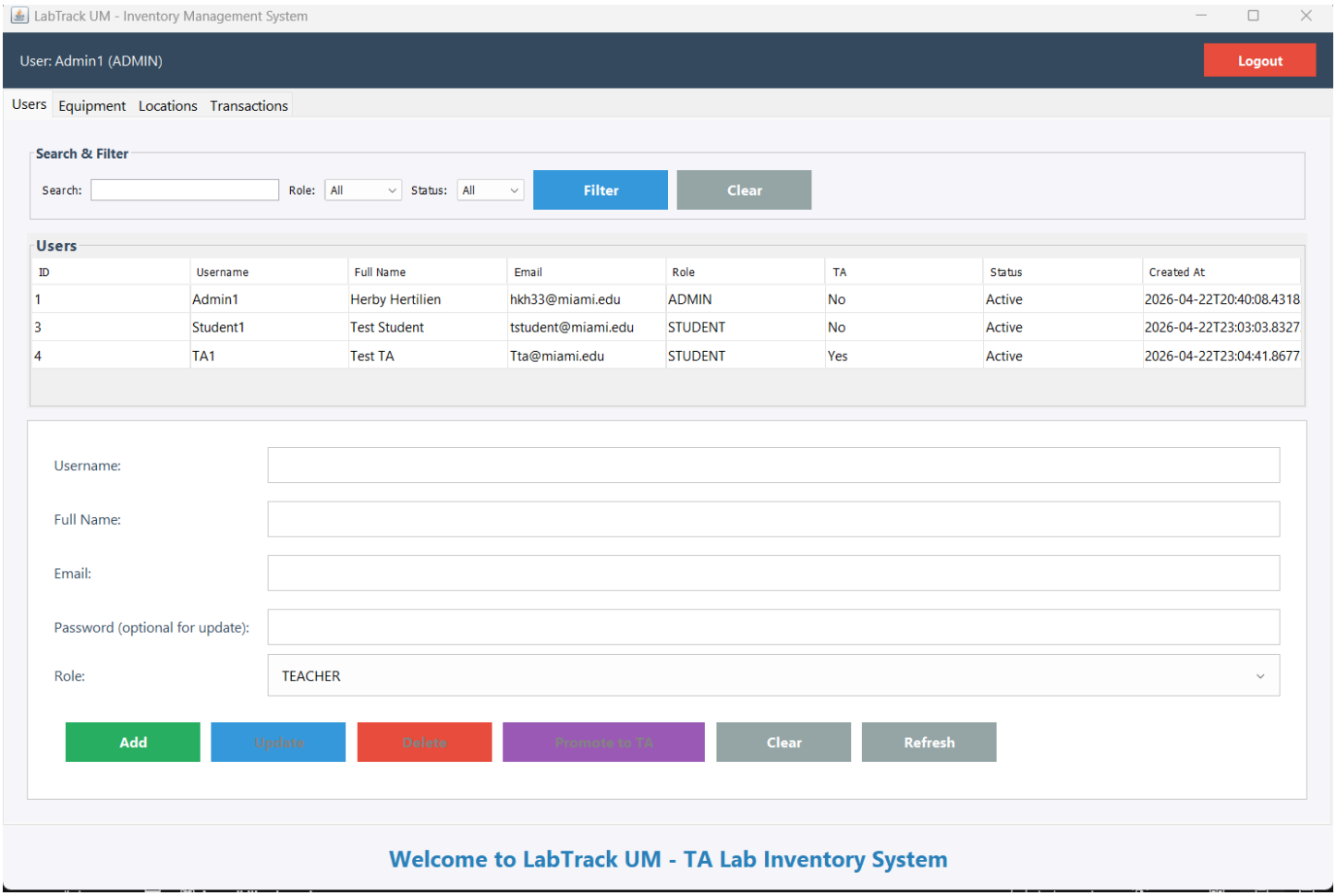


id	action	timestamp	equipment_id	user_id
1	CHECKOUT	2026-04-23 03:03:13.637288	1	3
2	CHECKIN	2026-04-23 03:03:28.733339	1	3
3	CHECKOUT	2026-04-23 03:06:31.951280	1	4
4	CHECKOUT	2026-04-23 03:11:44.943787	2	3
5	CHECKIN	2026-04-23 03:12:10.710765	2	3
6	CHECKOUT	2026-04-23 03:12:17.841636	2	3
7	CHECKIN	2026-04-23 03:12:25.812746	2	3
NULL	NULL	NULL	NULL	NULL

4. UI Testing

- Navigated between panels using MainFrame
- Tested buttons and forms
- Verified data displays correctly in UI

Expected: UI responds correctly and updates data
Result: Passed



5. Issues Found

During testing, one issue was identified:

- Student users currently have too many permissions
- They are able to create and manage equipment and locations

This does not affect system functionality, but it should be restricted in a future update using role-based access control.

6. Conclusion

Testing confirmed that the system is fully functional and all major features work as expected. The application runs correctly, and the UI interacts properly with the backend. Minor improvements can be made to user permissions, but overall the system is ready for use.